LogicWhile Assessment Suite

FUNCTIONAL ENGINEERING SPECIFICATION • GATE MODULE

Doc ID: AL-SPEC-2026-REG
Status: Approved Baseline
Target: Web & Tablet Engine

100%	4 Batches	5 Sections	< 200ms
IDENTITY UNIQUENESS	YOP: 2027 TO 2030	A, B, C, D, E MUTEX	CHECK & HANDSHAKE

1. EXECUTIVE OVERVIEW & FORM OBJECTIVES

The Candidate Registration Module serves as the high-throughput authentication gate for concurrent live assessments. It captures rigorous demographic and institutional tracking tokens necessary for generating multi-tier real-time leaderboards (College Rank, Department Rank, and Section Percentile).

2. MANDATORY DATA FIELD DICTIONARY

Field Name	UI Type	Type / Rule	Allowed Values & Validation Pattern	System Role
First Name	TEXT INPUT	REQUIRED	Min 2 characters; Alphabetic only; Auto-capitalizes first letter.	Public Leaderboard Display
Last Name	TEXT INPUT	REQUIRED	Min 1 character; Alphabetic only; Auto-capitalizes first letter.	Official Scorecard Record
College	RADIO DROPDOWN	REQUIRED	Searchable combobox with select list + radio buttons for top campuses.	Institution Isolation Key
Year of Passing	RADIO GROUP	REQUIRED	Strictly discrete values: **2027**, **2028**, **2029**, **2030**.	Cohort Normalization
Roll Number	TEXT INPUT	REQUIRED	Alphanumeric (e.g., 22B91A0501). Auto-uppercase. Uniqueness checked.	Primary Unique Index
Branch Code	SELECT / PILL	REQUIRED	Restricted to: `CSE`, `IT`, `ECE`, `EEE`, `MECH`, `CIVIL`, `AIDS`.	Domain Standings Filter
Section	RADIO GROUP	REQUIRED	Single-selection radio group: **A**, **B**, **C**, **D**, **E**.	Classroom Micro-Rankings
Email ID	EMAIL INPUT	REQUIRED	RFC 5322 standard email address format. Domain validation enabled.	Credential & Result Delivery
Contact Number	TEL INPUT	REQUIRED	10-digit mobile number matching `^[6-9][0-9]{9}$`.	Critical SMS Notifications

> **Data Integrity Invariant**
> The system strictly enforces composite uniqueness on `(college_id, roll_number, exam_session_id)`. Re-registration attempts with duplicate roll numbers inside the same testing session will trigger an automatic credentials challenge rather than overwriting existing records.

3. VISUAL INTERFACE & COMPONENT HIERARCHY

The onboarding viewport organizes personal, academic, and contact vectors into progressive logical groupings, preventing visual fatigue before the competitive timer starts.

• LIVE ASSESSMENT ENTRANCE GATE • SYSTEM CHECK: WEBSOCKET READY

FIRST NAME * Srinivas

LAST NAME * Kumar

COLLEGE AFFILIATION (DROPDOWN + RADIO SELECTION) *

[▾] Select or search institution name...

◉ University College of Eng. ○ Tech Campus North ○ V.V. Engineering Inst. ○ Other Campus

YEAR OF PASSING (BATCH) *

○ 2027 ◉ 2028 ○ 2029 ○ 2030

SECTION SELECTION *

○ A ◉ B ○ C ○ D ○ E

ROLL / HALL TICKET NUMBER * 22B91A0501

BRANCH CODE * [▾] CSE (Computer Science)

EMAIL ID (SCORECARD RECIPIENT) * srinivas.k@college.edu

CONTACT NUMBER (WHATSAPP UPDATES) * +91 9876543210

4. RELATIONAL PERSISTENCE & JSON EXCHANGE FORMAT

```
POST /api/v1/assessment/register-candidate Content-Type: application/json { "first_name": "Srinivas", "last_name": "Kumar", "college_id": "COL-UNIV-042", "year_of_passing": 2028, "roll_number": "22B91A0501", "branch_code": "CSE", "section": "B", "email": "srinivas.k@college.edu", "contact_number": "9876543210", "timestamp": "2026-09-25T12:32:15+05:30" }
```

Submission Routing & Lobby Handoff

Upon successful HTTP 201 `Created` response, the client receives an encrypted JWT session claim and is routed directly to `/contest/:id/waiting-lobby` with an active WebSocket subscription to the room's Redis Sorted Set leader channel.